

ML algorithms text

- ① Bag of words ✓
- ② one hot encoding
- ③ TF-IDF encoding
- ④ N-Gram Representation
- ⑤ Count vectorizer

4 documents:

$D_1 =$ "The boy and the dog"

$D_2 =$ "The cat and the dog"

$D_3 =$ "Andrew is a tall boy"

$D_4 =$ "Ram is a good boy. Ratna is also good"

① Bag of words:

$D_1 =$ "the boy and the dog" $\rightarrow$ 'the', 'boy', 'and', 'the', 'dog'

$D_2 =$ The, cat, and, the, dog

$D_3 =$ Andrew, is, tall, boy

$D_4 =$ Ram, is, good, boy, Ratna, is, also, good.

creates vocabulary (CORPUS)

[also, and, andrew, boy, cat, dog, good, is, ram, ratna, tall, the] $\rightarrow$ 12 words tokens

	<u>the</u>	<u>boy</u>	<u>and</u>	<u>the</u>	<u>dog</u>
was					
also					
and					
andrew					
boy					
cat					
dog					
good					
is					
ram					
ratna					

is 0
 ram 0
 rating 0
 tall the 0
 the 2

→ $D_1 = [0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 2]$

$D_2 \rightarrow$ the cat and the dog

$D_2 = [, 2$

one hot encoding:

D_1, D_2

{and boy cat dog is the}

and $[1, 0, 0, 0, 0, 0]$
 boy $[0, 1, 0, 0, 0, 0]$
 cat $[0, 0, 1, 0, 0, 0]$
 dog $[0, 0, 0, 1, 0, 0]$
 is $[0, 0, 0, 0, 1, 0]$
 the $[0, 0, 0, 0, 0, 1]$

$S_0 = [0, 1, 0, 0, 0, 0]$

and = 0

boy = 1

cat = 0

dog = 0

is = 0

the = 0

$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} = \text{remap}(-1, 1)$

one-hot encode

TF-IDF:

TF-IDF (stop-words = 'length')

The, and, also, is, a,

$D_1 \rightarrow$ the boy and the dog

$D_1 \rightarrow$ boy, dog | $D_2 =$ andrew tall boy
 - the, and

$D_1 \rightarrow \underline{\text{boy, dog}} \mid D_2 = \underline{\text{andrew tall boy}}$
 $D_2 \rightarrow \underline{\text{cat, dog}} \mid D_4 \equiv \underline{\text{ram good, ratna, good boy}}$
 $[\underline{\text{andrew, boy, cat, dog, good, ram, ratna, tall}}]$

word	Appears in doc	Frequency
<u>andrew</u>	D_3	1
<u>boy</u>	D_1, D_3, D_4	3
<u>cat</u>	D_2	1
<u>dog</u>	D_1, D_2	2
<u>good</u>	D_4	1
<u>ram</u>	D_4	1
<u>ratna</u>	D_4	1
<u>tall</u>	D_3	1

$N = 4$

$$IDF(t) = \log \left(\frac{1+N}{1+DF(t)} \right) + 1$$

①

$$\underline{\text{andrew}} = \underline{\log \left(\frac{1+4}{1+1} \right) + 1} = \log \left(\sqrt{2} \right) + 1$$

$$= \log 2 \cdot 1 + 1$$

$$= \underline{0.9163} + 1 = 1.9163$$

andrew $\rightarrow 1.9163$

cat $\rightarrow "$

good $\rightarrow "$

ram $\rightarrow "$

ratna $\rightarrow "$

tall $\rightarrow "$

if $\rightarrow$ dog $\rightarrow \log \left(\frac{1+4}{1+2} \right) + 1$

$$= \log \left(\sqrt{3} \right) + 1 = \log (0.16) + 1$$

$$=$$

cont IDF

boy	1.22	Count IDF
dog	1.501	1.22
		1.5106

$$D, [0, \underline{1.23}, 0, \underline{1.501}, 0, 0, 0, 0]$$

$$D_2 = \frac{\text{TFIDF}}{\sqrt{\sum \text{IDF}^2}}$$

$$\text{boy} = 1.22, \text{dog} = 1.501$$

$$= \sqrt{1.2^2 + 1.5^2} = \sqrt{\quad} = 1.9$$

$$\text{boy} = \frac{1.22}{1.9} = 0.6 \quad \text{dog} =$$

$$D_1 = \underline{\text{The, boy, and, the, cat}}$$

The boy,
boy and
and the
the cat,

$$D_2 =$$

$$(\text{The boy, boy and, and, the, the cat})$$

(The sun, boy and, and, the, the cat)

- ① count vectorize, NLU
- ② one hot
- ③ TF-IDF
- ④ N-Gram

Web scraping:

CSV
Excel
APIs
JSON data

process of automatically collecting data from

website.

Open webpage → Read HTML → Extract required data → store it

product name	price
iPhone 13	1,20,000
Price	4.5
Rating	2/100
Review	8hr
Availability	

Unstructured

<h1> product name </h1>

 \$\$\$

<p class="rating"> 4.5 </p> XML

- 1 Choose website
- 2 Send HTTP Request
- 3 Download HTML
- 4 Parse HTML
- 5 Extract required data
- 6 Clean the data
- 7 Store in csv
- 8 Analyze data
- 9 Feature Extraction
- 10 apply algorithm

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